

Practice sheet: Natural and Step Responses of RL and RC Circuits

This sheet is not marked for correctness, and AI tools are allowed. The device-free quiz is built from twins of these questions, so make sure you can do each one on your own.

- **Two-tier items** ask for an answer *and* a reason. Get both right; on the quiz each earns half the marks.
- **Predict first:** where asked, write your prediction before you calculate.
- **Confidence:** next to each answer, write S (sure) or U (unsure). This is not marked; it shows you what to revise.
- **Conventions:** “a long time” means the circuit has reached dc steady state: an inductor is a short circuit and a capacitor an open circuit. Every response here has the form $x(t) = x_f + (x_0 - x_f)\exp(-t/\tau)$, with $\tau = L/R$ or $\tau = RC$, where R is the resistance the inductor or capacitor sees after switching. Check every answer: $x(0)$ must equal the initial value, $x(\infty)$ the final value, and τ must be positive.

S1. What cannot jump? (two-tier) A switch moves at $t = 0$ in a circuit with a resistor and an inductor. Which of these must always have the same value just after the switch moves ($t = 0^+$) as just before ($t = 0^-$), in any circuit of this kind?

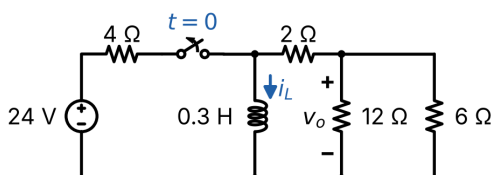
- | | |
|---------------------------|---------------------------|
| a) The inductor's voltage | b) The inductor's current |
| c) The resistor's current | d) The resistor's voltage |

Reason:

- An inductor's current cannot change instantly: a jump would need an infinite voltage.
- An inductor's voltage cannot change instantly, because its current is continuous.
- A resistor's current cannot jump, because Ohm's law fixes it.
- A resistor's voltage cannot jump, because the resistance does not change.

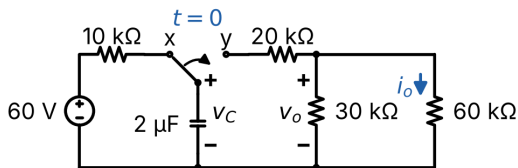
Follow-up: name the quantity that cannot jump in a circuit with a capacitor.

S2. RL natural response. As in Nilsson Example 7.1. The switch has been closed for a long time and opens at $t = 0$.



- Find $i_L(0)$.
- Find the time constant τ .
- Find $i_L(t)$ for $t \geq 0$ and $v_o(t)$ for $t \geq 0^+$.
- Find $v_o(0^-)$ and $v_o(0^+)$. Why can v_o jump when i_L cannot?
- What percentage of the initial stored energy is dissipated in the $12\ \Omega$ resistor?

S3. RC natural response. As in Nilsson Example 7.3. The switch has been in position x for a long time and moves to position y at $t = 0$.



- Find $v_C(0)$.
- Find τ .
- Find $v_C(t)$ for $t \geq 0$, and $v_o(t)$ and the current i_o down through the $60 \text{ k}\Omega$ for $t \geq 0^+$.
- How much energy is dissipated in the $60 \text{ k}\Omega$ resistor?

S4. One time constant later (two-tier). In a natural response, what fraction of the initial value remains after one time constant?

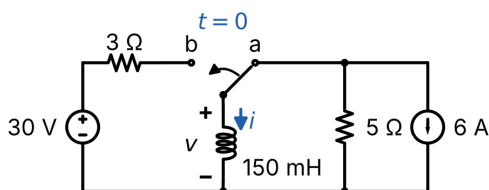
- 63%
- 0
- 37%
- 50%

Reason:

- The time constant is the time the response takes to reach its final value.
- After one time constant, $x = x_0 e^{-1} = 0.37x_0$.
- Half of what remains decays in each time constant.
- After one time constant, $1 - e^{-1} = 63\%$ of x_0 remains.

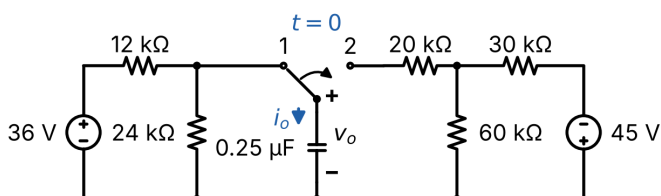
Follow-up: after how many time constants is less than 1% left?

S5. RL step response. As in Nilsson Example 7.5. The switch has been in position a for a long time. At $t = 0$ it moves to position b; it makes contact at b before it breaks contact at a, so the inductor current is never interrupted.



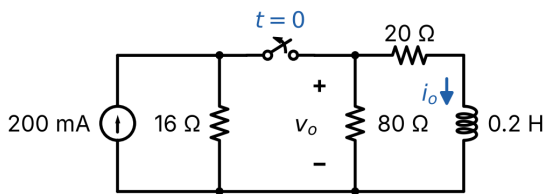
- Find $i(0)$, $i(\infty)$ and τ .
- Find $i(t)$ for $t \geq 0$.
- Find $v(t)$ for $t \geq 0^+$. Why is $v(0^+)$ larger than the 30 V of the source?
- When does v equal 30 V?

S6. RC step response with a Thévenin equivalent. As in Nilsson Example 7.6. The switch has been in position 1 for a long time and moves to position 2 at $t = 0$.



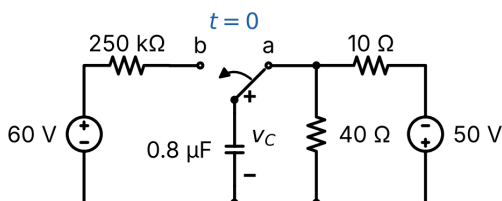
- Find $v_o(0)$.
- For $t \geq 0$, find the Thévenin equivalent that the capacitor sees.
- Find $v_o(t)$ for $t \geq 0$.
- Find the capacitor current i_o , referenced into its + terminal, for $t \geq 0^+$.

S7. The general method, RL. As in Nilsson Example 7.7. The switch has been closed for a long time and opens at $t = 0$.



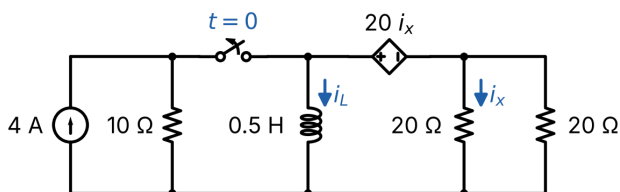
- Find $i_o(0)$.
- Find τ .
- Find $i_o(t)$ for $t \geq 0$.
- Find $v_o(0^-)$, $v_o(0^+)$ and $v_o(t)$ for $t \geq 0^+$.

S8. The general method, RC from a negative voltage. As in Nilsson Example 7.8. The switch has been in position a for a long time and moves to position b at $t = 0$.



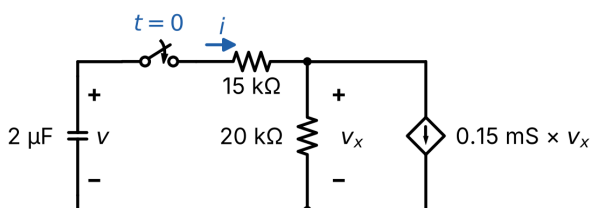
- Find $v_C(0)$. Watch the polarity.
- Find $v_C(t)$ for $t \geq 0$.
- Find the capacitor current i , referenced into its + terminal, for $t \geq 0^+$.
- When is v_C zero?

S9. A dependent source sets τ . As in booklet Problem 7.7. The switch has been closed for a long time and opens at $t = 0$.



- Find $i_L(0)$.
- For $t \geq 0$, find the resistance the inductor sees, using a test source.
- Find τ and $i_L(t)$ for $t \geq 0$.
- Find $i_x(t)$ for $t \geq 0^+$.

S10. A dependent source in an RC circuit. The $2 \mu\text{F}$ capacitor has been charged to 30 V. At $t = 0$ the switch closes.



- Find the resistance the capacitor sees, using a test source.
- Find τ and $v(t)$ for $t \geq 0$.
- Find $i(t)$ and $v_x(t)$ for $t \geq 0^+$.

S11. Spot the error. The worked solution below contains the kind of polarity slip AI chat tools often make. For $t < 0$ a $0.4 \mu\text{F}$ capacitor has been connected for a long time across the $24 \text{ k}\Omega$ resistor of a divider: a 36 V source in series with $12 \text{ k}\Omega$ and $24 \text{ k}\Omega$. The source's $+$ terminal connects to the bottom of the $24 \text{ k}\Omega$, so the top of the $24 \text{ k}\Omega$ is negative. At $t = 0$ the capacitor is switched to a 48 V source in series with $125 \text{ k}\Omega$. v_C is positive at the top.

1. $\tau = (125 \text{ k}\Omega)(0.4 \mu\text{F}) = 50 \text{ ms}$.
2. $v_C(0) = 36 \times 24/(12 + 24) = 24 \text{ V}$.
3. $v_C(\infty) = 48 \text{ V}$.
4. $v_C(t) = 48 + (24 - 48)e^{-20t} = 48 - 24e^{-20t} \text{ V}$.
5. Check: $v_C(0) = 24 \text{ V}$ and $v_C(\infty) = 48 \text{ V}$, as found.

- a) Which is the first wrong line?
- b) Correct it and find $v_C(t)$.
- c) Line 5 calls itself a check. Why did it not catch the error?
- d) When is v_C zero?

S12. Spot the error in a time constant. An uncharged $0.4 \mu\text{F}$ capacitor is connected at $t = 0$ through $5 \text{ k}\Omega$ to a node N. Node N connects through $30 \text{ k}\Omega$ to a 60 V source and through $60 \text{ k}\Omega$ to ground.

1. $v(0) = 0$, since the capacitor is uncharged.
2. $v(\infty) = 60 \times 60/(30 + 60) = 40 \text{ V}$.
3. $\tau = (5 + 30 + 60 \text{ k}\Omega)(0.4 \mu\text{F}) = 38 \text{ ms}$.
4. $v(t) = 40 - 40 \exp(-t/0.038) \text{ V}$.

- a) Which is the first wrong line?
- b) Find the correct τ and $v(t)$.
- c) What resistance does the capacitor see, and how do you find it?

S13. Work backwards. A capacitor charges through $10 \text{ k}\Omega$. Its voltage is 10 V at $t = 0$, rises toward 40 V , and reads 30 V at $t = 2 \text{ ms}$.

- a) Find τ .
- b) Find C .
- c) When does the voltage reach 39 V ?
- d) The same steps work for an inductor: a current rises from 0 toward 2 A through 20Ω and reads 1.5 A at $t = 6.93 \text{ ms}$. Find τ and L .

S14. An artificial pacemaker. As in Nilsson's Practical Perspective for this chapter. A source V_s charges a capacitor C through a resistor R , starting from 0 V . When v_C reaches $0.75V_s$, a controller discharges the capacitor in a negligible time and sends a pulse to the heart; then charging starts again.

- a) Show that each charging takes $t = RC \ln 4$, so the heart rate is $H = 60/(RC \ln 4)$ beats per minute.
- b) With $C = 2 \mu\text{F}$, what R gives 60 beats per minute?
- c) With $C = 2 \mu\text{F}$ and $R = 300 \text{ k}\Omega$, what is the heart rate?
- d) *Predict first:* what happens to the heart rate if R is doubled?

S15. How fast does a relay let go? A 0.5 H relay coil with 60 Ω winding resistance carries 0.2 A from a 12 V supply. When the driving switch opens at $t = 0$, the coil current continues through a diode (ideal: 0 V when conducting) in series with a 140 Ω resistor, connected across the coil.

- Find τ for $t \geq 0$.
- The relay releases when the current falls below 50 mA. When does it release?
- What voltage appears across the coil's terminals at $t = 0^+$? Take it positive at the terminal the supply made positive.
- Repeat (a) and (b) with the diode alone (no 140 Ω). Why do designers sometimes add the resistor?

S16. Check an answer without solving again. A 0.4 H inductor carries $i(0) = -2$ A. At $t = 0$ it is switched across a 20 V source in series with 5 Ω ; i is referenced in the direction the source drives current. Student 1: $i = 4 - 6 \exp(-12.5t)$ A. Student 2: $i = 4 - 2 \exp(-12.5t)$ A. Use $i(0)$, $i(\infty)$ and τ to decide which student is right.